JAVA PROGRAMMING
Common to CSE&IT

	L	T	P	C
IV Semester	**L**	**T**	**P**	**C**
Course Code:201CS4T07	3	0	0	3

Course Outcomes:

At the end of the Course, Student will be able to:

CO1: Apply Java features for problem solving.
CO2: Build applications using principles of OOPs, interfaces and Packages.
CO3: Develop programs using Exception Handling to handle run-time errors.
CO4: Develop applications using multithreading for inter thread communication.
CO5: Build JDBC applications for performing CRUD operations using MySQL.

Mapping of Course Outcomes with Program Outcomes:

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	-	-	-	-	-	-	-	-	-
CO2	2	2	2	-	-	-	-	-	-	-	-	-
CO3	2	2	2	1	-	-	-	-	-	-	-	-
CO4	3	2	3	-	-	-	-	-	-	-	-	-
CO5	2	2	3	-	1	-	-	-	-	-	-	2

Mapping of Course Outcomes with Program Specific Outcomes:

CO/PSO	PSO1	PSO2
CO1	1	-
CO2	2	-
CO3	1	-
CO4	1	-
CO5	2	-

Unit – I
Introduction to Java: History of Java, Java Features, Program Structure, Command Line Arguments, User Input to Programs. Building Blocks of Java: Identifiers, Data types, Literal Constants, Variables and its Scope, Formatted Output with printf() Method, Operators, Precedence and Associativity of Operators, Type Casting. Control Statements: Selection Statements: if-else, switch, Iteration Statements: while, do-while, for, for each, Transfer Statements: Break, Continue

Unit – II
Arrays: Introduction, Declaration and Initialization of Arrays, Accessing Elements of Array. Operations on Array Elements, Class Arrays, Two-dimensional Arrays, Arrays of Varying Lengths, Multi-dimensional arrays. Classes, Objects and Methods: Class Declaration, Creating Objects,Assigning One Object to Another, Methods, Constructors, this keyword, static keyword, final keyword, garbage collector, Access Control, Method Overloading, Constructor Overloading, Parameter Passing, Nested Classes. String Handling: StringClass, Methods for Extracting Characters from Strings, Methods for Comparison of Strings, Methods for Searching Strings, Methods for Modifying Strings, String Buffer Class and its methods, Class String Builder.

Unit – III
Inheritance: Inheritance, Types of Inheritance, Constructor Method and Inheritance, Super keyword, Method Overriding, Dynamic Method Dispatch, Inhibiting Inheritance of Class Using Final, Abstract Classes.Interfaces- Defining an interface, Implementing interfaces through classes, Multiple inheritance through interfaces.

Unit – IV

Packages: Introduction, Defining Package, Importing Packages and Classes into Programs, Path and Class Path, Access Specifiers, java.lang package, Wrapper Classes. Exception Handling: Introduction, Importance of try, catch, throw, throws and finally block, Multiple Catch Clauses, Rethrowing Exception, Nested try and catch Blocks, Unchecked Exceptions, Checked Exceptions,Custom Exceptions.

Unit – V

Multithreading: Introduction, Thread Life Cycle, Creation of Threads, Thread Priorities, Thread Synchronization, Inter-thread Communication- Suspending, Resuming, and Stopping of Threads. Java Database Connectivity: Introduction, JDBC Architecture, Installing MySQL and MySQL Connector/J, JDBC Environment Setup, Establishing JDBC Database Connections, ResultSet Interface, Creating JDBC Application.

Text Books:

1. The Complete Reference Java, Herbert Schildt, 8th Edition, TMH, 2014.
2. Java one step ahead, Anita seth, B.L.Juneja, First Edition, Oxford, 2017.

Reference Books:

1. .Introduction to java programming, by Y Daniel Liang, Seventh Edition, Pearson, 2017.
2. Core Java: An Integrated Approach, R.Nageswara Rao, Dream tech press, 2008.
3. Thinking in Java – Bruce Eckel, Fourth Edition, Prentice Hall, 2002.

Web Links:

1. https://nptel.ac.in/courses/106/105/106105191/
2. http://java.sun.com/docs/books/tutorial/
3. http://www.tutorialspoint.com/java
4. http://www.javatpoint.com
5. http://www.w3schools.com/java

JAVA PROGRAMMING LAB
Common to CSE&IT

	L	T	P	C
IV Semester				
Course Code:201CS4L05	0	0	0	1.5

Course Outcomes:

At the end of the Course, Student will be able to:

CO1: Apply class, inheritance, and interface for problem solving.
CO2: Build applications using packages to group liked classes.
CO3: Develop error free programs using exception handling.
CO4: Develop a solution for ITC using multithreading.
CO5: Apply event handling for interactive applications.
CO6: Develop JDBC applications for performing CRUD operations using MySQL.

Mapping of Course Outcomes with Program Outcomes:

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	1	-	-	-	-	1	2	-	2
CO2	2	2	2	1	-	-	-	-	1	2	-	2
CO3	2	3	3	1	-	-	-	-	1	2	-	2
CO4	2	2	3	1	-	-	-	-	1	2	-	2
CO5	2	2	3	1	-	-	-	-	1	2	-	2
CO6	2	2	3	1	2	-	-	-	1	2	-	2

Mapping of Course Outcomes with Program Specific Outcomes:

CO/PSO	PSO1	PSO2
CO1	-	1
CO2	-	1
CO3	-	1
CO4	-	1
CO5	-	2
CO6	-	2

List of Experiments:

Week – 1
1) Basic Programs
1.1)Write a Java program to display default value of all primitive data type of JAVA
1.2)Write a Java program to find the discriminant value D and find out the roots of the quadratic equation of the form ax2+bx+c=0.
1.3)Five Bikers Compete in a race such that they drive at a constant speed which may or may not be the same as the other. To qualify the race, the speed of a racer must be more than the average speed of all 5 racers. Take as input the speed of each racer and print back the speed of qualifying racers.

Week – 2
2) Control Flow Statements
2.1)Write a Java program to select all the prime numbers within the range of 1 to 100.
2.2)Write a Java program to Find the sum of all even terms in the Fibonacci sequence up to the given range N.
2.3)Write a Java program to check whether a given number is Armstrong or not.

Week – 3
3) Arrays
3.1)Write a Java program to implement binary search.
3.2)Write a Java program to sort for an element in a given list of elements using bubble sort.
3.3)Write a Java program to sort for an element in a given list of elements using merge sort.

Week – 4
4) Class Mechanism
4.1)Write a Java program to display the details of a person. Personal details should be given in one method and the qualification details in another method.
4.2)Write a Java program to implement constructor and constructor overloading.
4.3)Write a Java program to implement method overloading.

Week – 5
5) Strings
5.1)Write a Java program to sort given set of strings.
5.2)Write a Java program for using String Buffer to remove or delete a character.

Week – 6
6) Inheritance
6.1)Write a Java program to implement Single Inheritance.
6.2)Write a Java program to implement multi level Inheritance.
6.3)Write a Java program to find the areas of different shapes using abstract classes.

Week – 7
7) Inheritance-continued
7.1)Write a Java program for "super" keyword.
7.2)Take the details of internal exam marks in one Interface. Take the details of external exam marks in another interface. Write a Java program to find the total marks obtained in each subject by a student. (Note: Make use of Multiple Inheritance using interfaces.)
7.3)Write a JAVA program that implements Runtime polymorphism

Week – 8
8) Packages
8.1)Write a Java program that import and use user defined package.
8.2)Write a Java program to illustrate the use of protected members in a package.

Week – 9
9) Exception Handling
9.1)Write a Java program to illustrate exception handling mechanism using multiple catch clauses.
9.2)Write a Java program to make use of Built-in and user-defined Exceptions in handling a run time exception.

Week – 10
10) Multithreading
10.1)Write a Java program that creates threads by extending Thread class .First thread display "Good Morning "every 1 sec, the second thread displays "Hello "every 2 seconds and the third display "Welcome" every 3 seconds, (Repeat the same by implementing Runnable).
10.2)Write a Java program to solve Producer-Consumer problem using synchronization.

Week – 11
11) Event Handling
11.1) Write a Java program to illustrate the Keyboard Events by using an applet code
11.2) Write a Java program to illustrate the Mouse Events by using an applet code.
11.3) Write a Java program to generate a simple calculator using AWT components.

Week – 12
12)Write a JDBC program to perform the following operations by connecting to MYSQL database.
12.1)Inserting Data into Table
12.2)Updating Data in the Table.
12.3)Deleting Data From the Table based on a column value.

List of Augmented Experiments:

13)Create an interface which consists of methods called no of watts consumabl, luminescent value, efficiency in percentage. Write classes for different categories of bulbs like LED, tube light and find out which light is efficient in terms of consumption.

14)Write a Java program that simulates a traffic light. The program lets the user select one of three lights: red, yellow, or green with radio buttons. On selecting a button, an appropriate message with "Stop" or "Ready" or "Go" should appear above the buttons in selected color.

15)Write a Javaprogram to display analog clock using Applet.

16)Develop an applet that receives an integer in one text field, and computes its factorial Value and returns it in another text field, when the button named “Compute” is clicked.

Reference Books:

1. Java How to Program, H.M.Dietel and P.J.Dietel, Pearson Education/PHI, SixthEdition 2007.
2. Core Java: An Integrated Approach–R.Nageswara Rao, First Edition, JohnWiley and Sons Inc.,2015.
3. Java Tutorial: A Short Note on Basics-Sharon Biocca Zakhour, Soumya Kannan, Raymond Gallardo–Fifth Edition, Oracle Corp,2012.
4. Object Oriented Programming using Java–Simon Kendal, First Edition, 2009.
5. Java:The fundamentals of Objects and Classes– DavidEtheridge,FirstEdition,2009.

Web Links:

1. http://www.programmingtutorials.com/java.aspx
2. http://www.javacodegeeks.com
3. http://java.sun.com/developer/onlineTraining/
4. http://java.sun.com/learning
5. http://www.kodejava.org